Workpath SDK User Guide

Product Group: HP Workpath SDK
Version: 1.6.3
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1 Introduction

This document provides information for users of the HP Workpath SDK.

1.1 Document conventions

The following conventions are used throughout this document to define, describe, and discuss the API: function prototypes, data structure definitions, enumerated constants.

- **Name** – Bold signifies item names such as button names and menu items.
- <name> – Angle brackets mark URLs.
- {value} – Curly braces represent values to be replaced.
- Identifiers for defined constants use all UPPERCASE letters.
- *Name* – Italicized type refers to formal function parameters, data structure types and fields, and other defined or enumerated types in the text body (for example, *streamtype*).
- Numbers are in decimal format unless otherwise noted.
- `Name` - Monospaced type is used for code fragments, function prototypes and data type definitions.
- Function names are usually accompanied by parentheses (for example, `function()`).

2 Overview

The **HP Workpath SDK** provides a comprehensive suite of features to allow application developers to easily integrate scanning, printing, configuring and device control functionalities in their applications.

The **HP Workpath SDK** comes preloaded with documentation, samples, and tool to jump start application development, and is geared for developers with no prior experience for HP devices.

2.1 Background

The HP Workpath SDK is designed to overcome challenges faced by developers seeking to integrate functionality for MFP devices into their applications.

These challenges are due to:

- variance in standard protocols used (Postscript, PCL, property PCL) across devices.
- variance in the architecture, protocols, and internals of MFP devices.
- difficulty debugging printer functionality due to lack of simulator support, requirement to learn company/model specific protocols and/or debugging tools.
- lack of control over device functions and settings, such as file formats used, duplex options, size and resolution, ability to check jobs in progress or accounting details.

The HP Workpath SDK solves these problems by introducing APIs, which are extremely simple and easy to use, even for developers with little or no experience with printer functionalities.

2.2 About this guide

This overview will explain:

- general architecture of the HP Workpath SDK
- layout of the HP Workpath SDK
- how to install and use the sample included with the HP Workpath SDK

2.3 HP Workpath SDK 1.6.3

The HP Workpath SDK V1.6.3 is the “next release” of the Workpath SDK V1.6.2. All apis and samples of V1.6.3 are compatible with V1.6.2.

2.4 Notice

2.4.1 How to migrate apps to Android12

Workpath solution is recommended to add ***android:exported =true*** or change TargetPlatform to 29.

See https://developer.android.com/about/versions/12/migration for details.

2.4.2 Notices using foreground services

Android default behavior is to keep the running process permanently if the app has foreground services. This is possible on a personal mobile device. In addition, this behavior requires the stored user privacy continuously.
However, Workpath supports multi users on single device. Only one user should run an app at a time. and all previous user's tasks should be terminated in secure.
Workpath enhances a platform action to terminate the app with the foreground service.
If reset/inactivity timeout/sign-out happens in Workpath device, Workpath framework clears the previous user information through clearing the recent task with foreground services. See "HP WorkpathSDK-forDevice-AccessoryOverview_v1.6.3.pdf" for examples.

3 HP Workpath SDK Structure

The HP Workpath SDK contains these key elements:

- **The Workpath SDK API** – The Workpath API is the WorkpathLib.aar file, and contains the abstracted functions used by applications to communicate with the HP device.
- **The Workpath SDK Service** – The Workpath SDK Service translates the wrapped code to the OXPd and specific protocols used by the device.

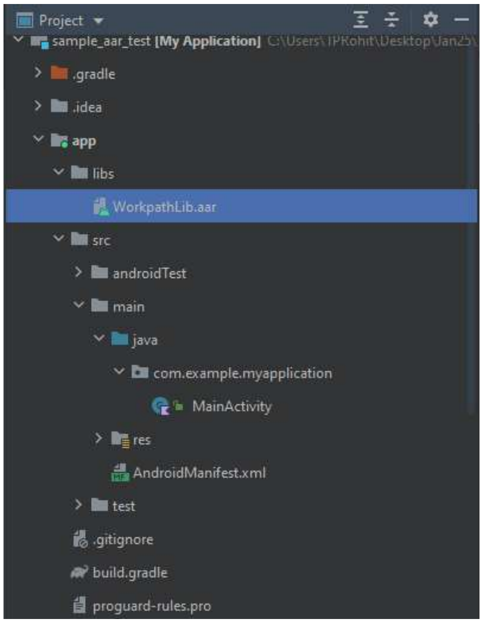
Figure 1. Key Components of the Workpath SDK

3.1 The Workpath SDK API

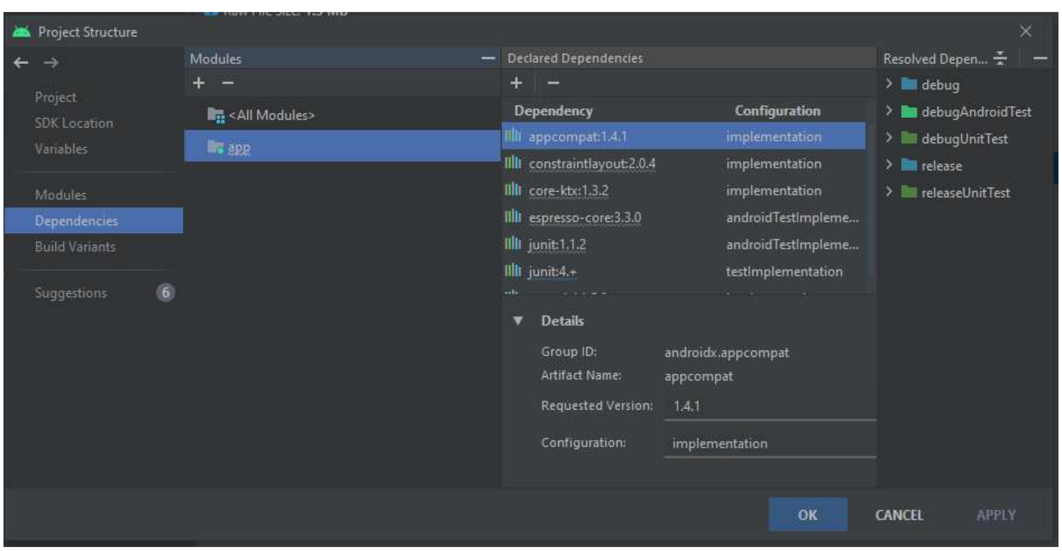
The Workpath SDK API is a collection of Java classes which should be used to communicate with Workpath SDK Service. To use the Workpath SDK API, import the Workpath SDK library (the WorkpathLib.aar is in Libs folder of the SDK) and compile the project by following the steps below:

Android Studio Flamingo | 2022.2.1 Patch 2

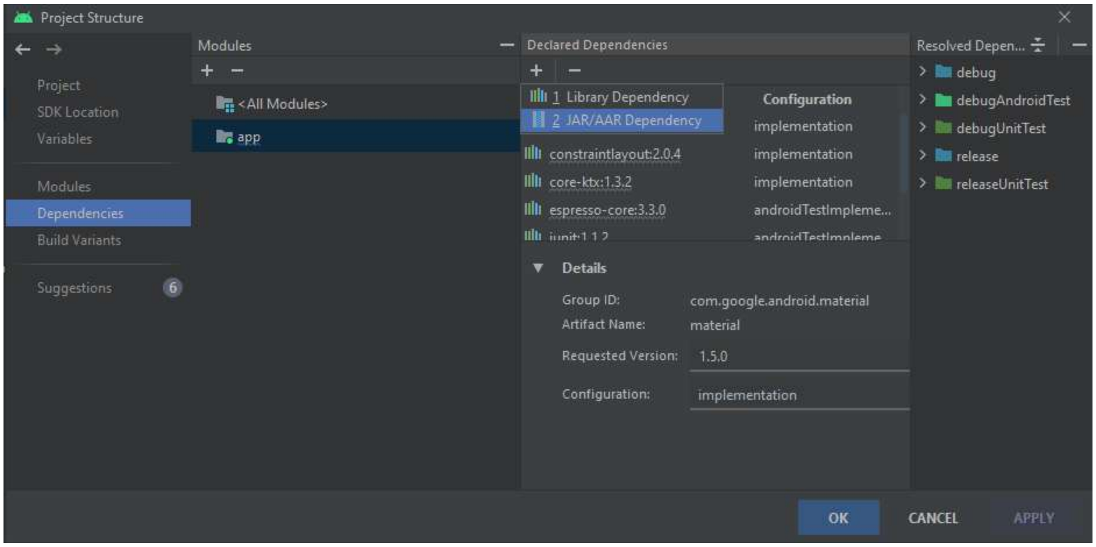
1. Select '**project**' type **>** paste the WorkpathLib.aar file in the libs folder of your project



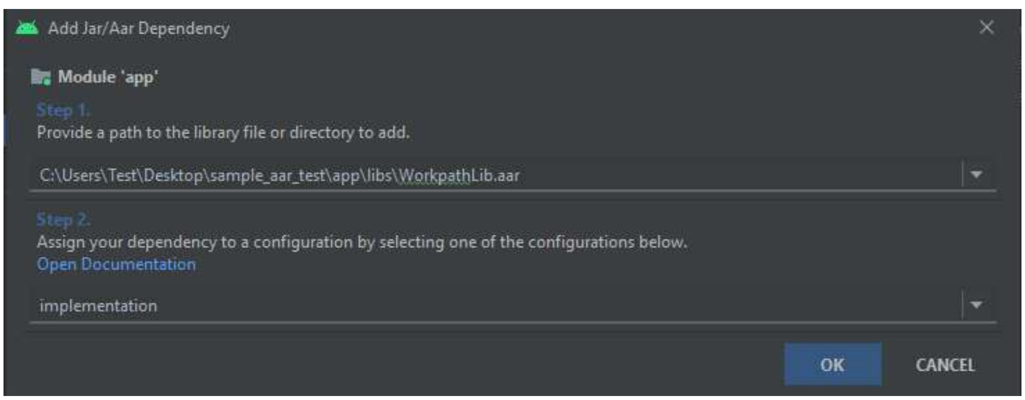
2. Right click on WorkpathLib.aar, click on **Copy path > Copy absolute path**.
3. Select **File > Project structure**.



4. Select **Declared Dependencies >** click on '**+**' icon **>** '**JAR/AAR dependency**'.



5. Paste the path of the AAR copied earlier, set configuration as 'implementation' and click OK.



6. Click **APPLY**, **OK,** and wait for Gradle sync to complete.

3.2 Workpath SDK Service

The HP Workpath SDK Service is preinstalled in the Workpath Platform compatible devices as a part of system application.

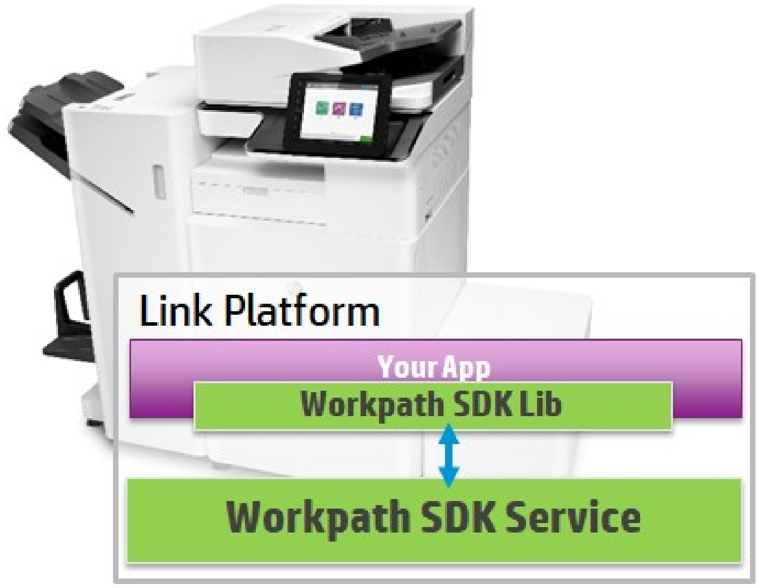


Figure 2. The HP Workpath SDK

3.3 Compatible devices

The HP Workpath SDK V1.6.3 is compatible with the HP Color LaserJet Flow, Laser Jet Flow and HP PageWide MFPs which support Workpath Platform introduced in FutureSmart 5.9.2.2 or higher version.

To get more details with compatible firmware, visit to the online: http://developers.hp.com

4 The HP Workpath SDK layout

APIDocs\	
\WorkpathLib-javadoc	The master documentation of all Workpath SDK API
WorkpathLib-javadoc.jar	Javadocs of Workpath SDK API for IDE
Libraries\	
WorkpathLib.aar	The Workpath SDK classes and helpers for interacting with Workpath SDK Services
Samples\	
\ExampleAPIServices	Collection of Java examples that exercise to use the methods of each Workpath SDK API
\sources	Source codes of samples
\AccessoryAgentSample	Source code of a complete user scenario for using Accessory api with authentication through the broadcast event
\AccessorySample	Source code of a complete user scenario for using Accessory api
\AccessoryServiceSample	Source code of a complete user scenario for using Accessory api with authentication through the service
\AccessSample	Source code of a complete user scenario for using AccessService api
\AttestationSample	Source code of a complete user scenario for using Attestation api
\AuthenticationAgent	Source code of a complete user scenario for using AuthenticationAgent in workpath
\AuthenticationAgentWithPrePrompt	Source code of a complete user scenario for using AuthenticationAgent with preprompt api in workpath
\AuthorizationSample	Source code of a complete user scenario for using Authorization api
\CopySample	Source code of a complete user scenario for using CopierService api
\DeviceInfoSample	Source code of a complete user scenario for using DeviceService api
\PrintSample	Source code of a complete user scenario for using PrinterService api
\ScanSample	Source code of a complete user scenario for using ScannerService api
\ConfigSample	Source code of a complete user scenario for using ConfigService api
\EmailSample	Source code of a complete user scenario for using Email api

\LauncherSample	Source code of a complete user scenario for using Launcher api
\MassStorageSample	Source code of a complete user scenario for using MassStorage api
\MultiLanguageSample	Source code of a complete user scenario for understanding multi-langueage in workpath
\DeviceEventSample	Source code of a complete user scenario for understanding Device Events in workpath
\DeviceUsageSample	Source code of a complete user scenario for understanding DeviceUsage Service in workpath
\StatisticsSample	Source code of a complete user scenario for understanding Statistics Service in workpath
\SuppliesSample	Source code of a complete user scenario for understanding Supplies Service in workpath
\WebServiceSample	Source code of a complete user scenario for understanding Web Service in workpath
\apks	Pre-built APK files to run Workpath SDK samples in Enhanced Debug Experience(EDX) Mode
\hpks	Pre-packaged HPK files to run Workpath SDK samples in Link Debug Bridge(LDB)
\ExampleExtensions	Collection of Java examples that integrate with Workpath platform
\sources	Source codes of samples
\GoogleSigninSample	Source code of a complete user scenario for integrating with Google Sign-in
Tools\	
\hpktool	HPK Tools and guide for creating and installing hpk file
Documentations\	The documentations of all Workpath SDK features and samples

5 SDK installation and setup

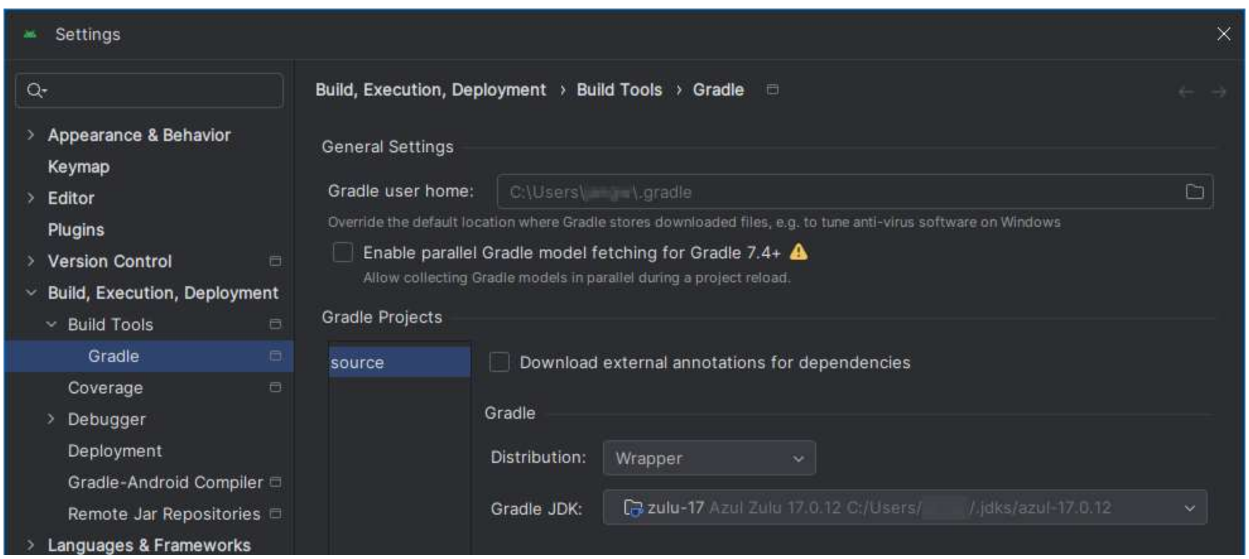
The following sections provide step-by-step setup instructions for installing the SDK and using the pre-built examples on multiple environments.

Build tools (gradle and Android SDK) and internet connection are required to download libraries for build and run the pre-built examples.

5.1 Development system requirements

Developing Workpath applications requires installation of the following prerequisites:

- **SDK**
    - MinSdkVersion API 26 / LatestSdkVersion API 31
- **IDE** (Integrated Development Environment):
    - The latest Android Studio and Android SDK. Refer to http://developer.android.com/sdk/index.html#Requirements to set up the development environment.
- **JDK**
    - JDK 17



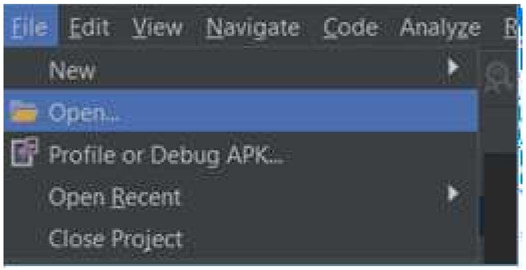
- **Gradle**
    - Android Gradle plugin version 7.4.2
    - Gradle version 8.0

5.2 Build, Install and test the SDK examples

5.2.1 Build the SDK examples

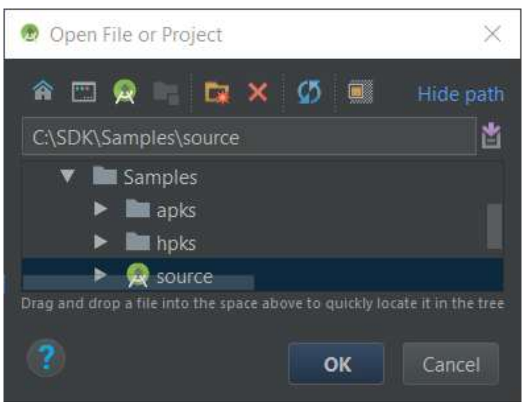
1. Open Android Studio.

2. Select the File drop-down menu, then select **Open…** to open the **Open File or Project**.



Open Project

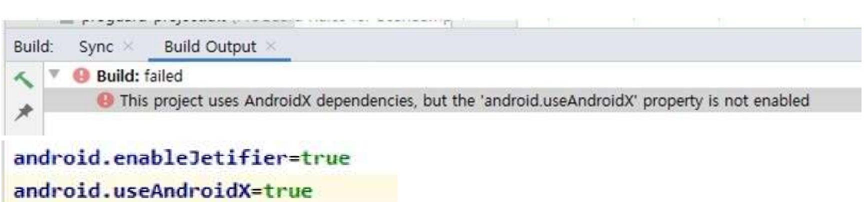
3. Select source to open samples.



Select Sample Project

4. Click **OK** to complete the open project, and wait for Gradle Sync completion.

NOTE: If there's build error as below, check "gradle.properties" and add 2 options under source directory.



5.2.2 Install the SDK examples

Step1. Enable Workpath platform on device.

Application is running by Workpath platform, so the first step is to enable Workpath platform on device's EWS.

Step2. Make a hpk file with HPK Tools.

Any app intended for use on any HP device must be specially packaged (.hpk), SDK provides a HPKTool to make an app to hpk file.

Step3(Development environment only). The Workpath Platform provides two ways to install your unsigned app on a device:

- Step3-1. Enabling the Link Debug Bridge and installing .hpk[NOTE] file using HPK Tool
- Step3-2. Enabling the Enhanced Debug Experience mode and installing .apk file using adb

Step3 is allowed for development environment. In customer environment, all applications must be signed by HP through verification and validation process (V&V) of App Center. Visit "HP site" to get more information of V&V.

NOTE: HPK is a package of a compressed archive using .ZIP file format where the files follow specific conventions. The primary purpose of a HPK package is to encapsulate a single App and meta-data associated with the App.

5.2.2.1 Step1. Enabling the Workpath Platform

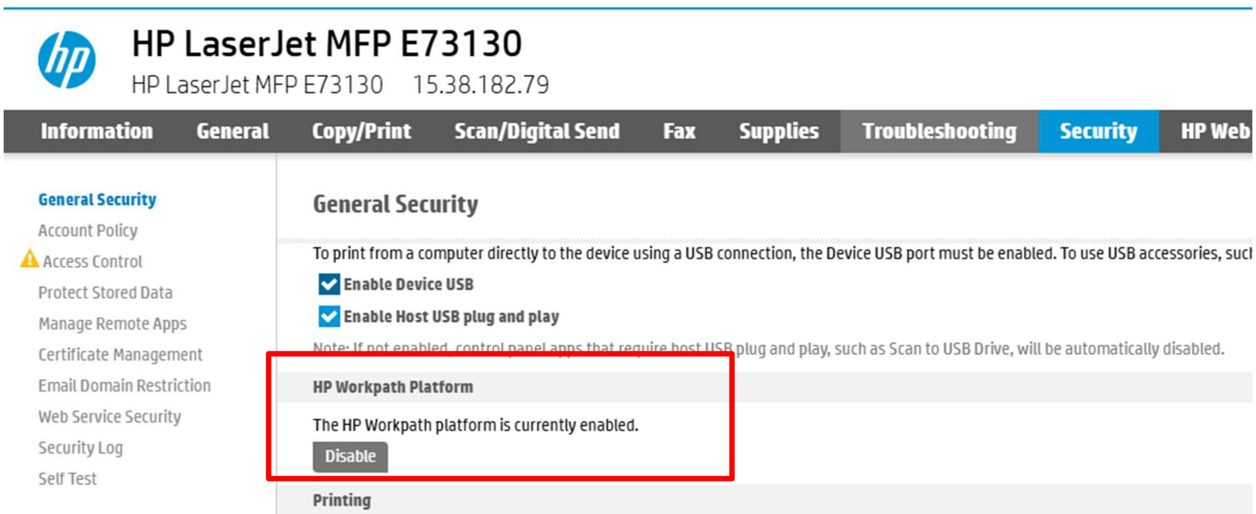
The Workpath platform is not enabled by default. To test your app, enable the mode. To enable it, follow these steps:

A. Check the list of compatible devices to be sure your device supports the Workpath platform.

B. Upgrade the device to the firmware.

C. Enable the Workpath Platform using the device's EWS.

1. Sign in
2. Navigate to the Security Tab
3. Navigate to the General Security page
4. Click on the Enable button in the HP Workpath Platform section of the page
5. Click Restart at the bottom of the displayed page

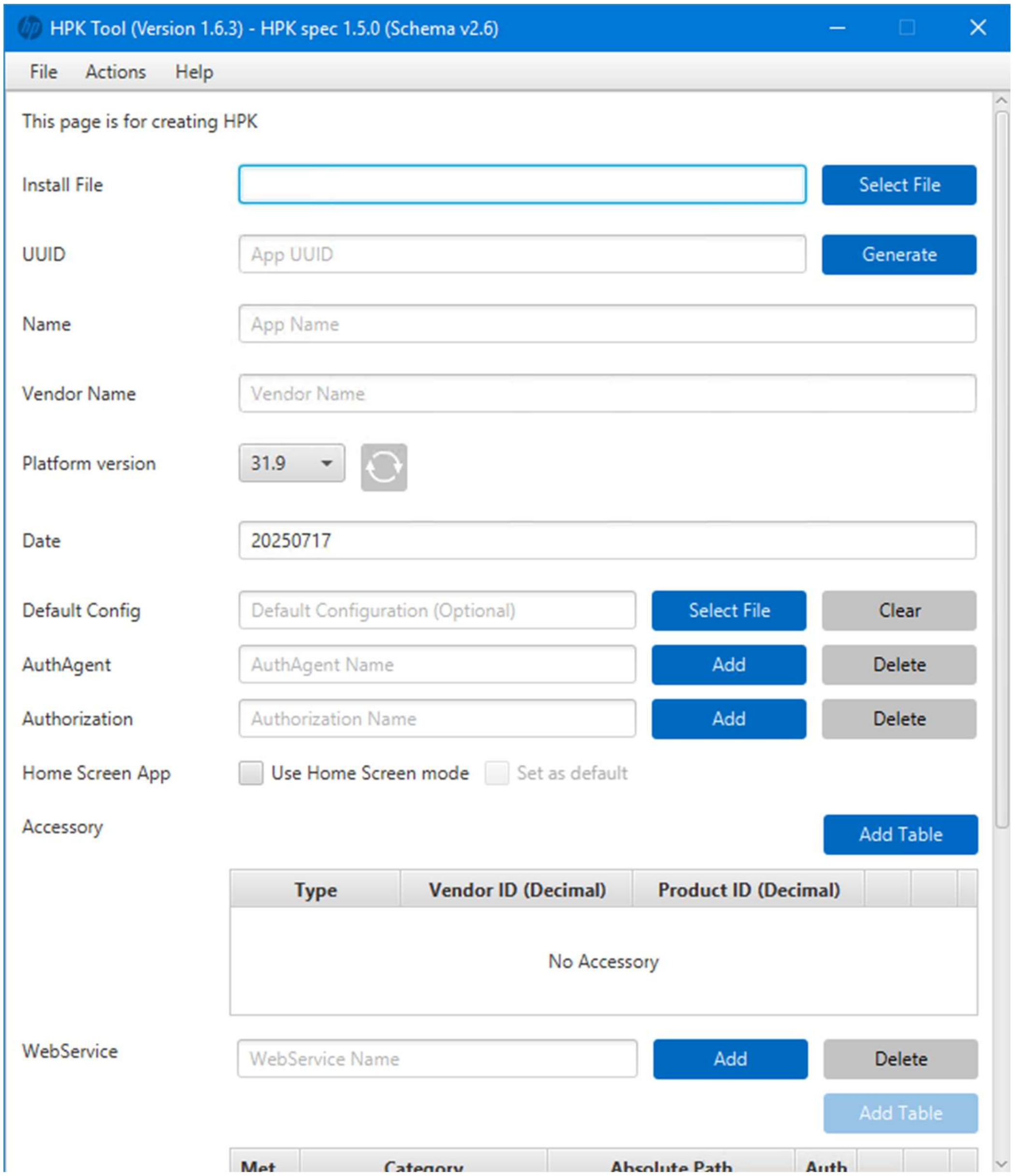


Menu to enable Workpath platform on EWS

5.2.2.2 Step2. Make a hpk file using HPK Tool.

Follow the steps below to create hpk file in a Windows:

1. Open the **HPKTool.exe** in **Tools > hpktool** folder of the SDK package.
2. In main dialog, select apk file and fill information.
3. Click **Create HPK** to create hpk file.



HPK Tool Pop-up for creating hpk file

5.2.2.3 Step3-1. Enabling the Link Debug Bridge and installing HPK file

The Link Debug Bridge (LDB) allows you to install and debug[note] your unsigned app on a device over the network using adb commands.

NOTE: To debug application, first, connect to a device.

adb connect <device_ip_address>

Then, run adb command for checking logs.

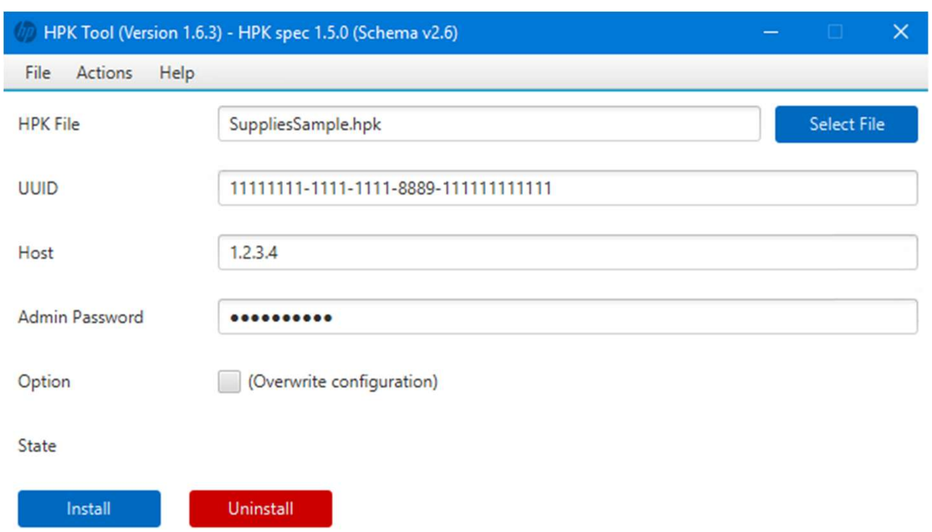
adb logcat

To enable the LDB and install unsigned hpk file, follow these steps:

1. Go to the device, then sign into the device as an administrator.
2. Open the **Settings** app.
3. Tap **Developer Options.**
4. Tap **HP Workpath Debug Bridge**, then tab **Enable**.
5. When prompted, enter your developers.hp.com account credentials (username and LDB Service Key[note]).

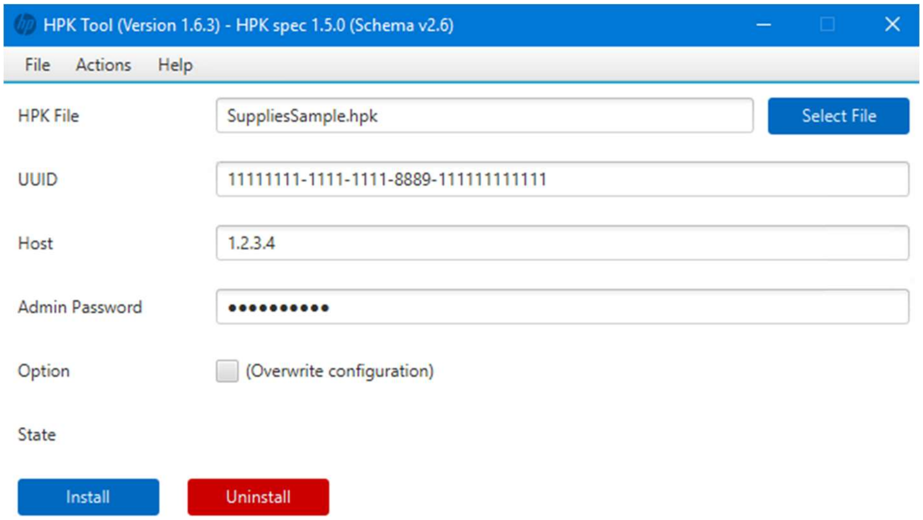
NOTE: LDB Service Key is personal password for the access Link Debug Bridge. User who has the account in "HP's Developer Portal(https://developers.hp.com)" can create the key after sign-in to the web.

6. If you change the mode, applications will be cleaned. Then, device will be restarted for initializing Workpath platform.
7. If device is restarted, open **HPKTool.exe** in **Tools > hpktool** folder of SDK package.
8. Select **Install/Uninstall .hpk** from the **Actions**, then select hpk file with **Select file**.



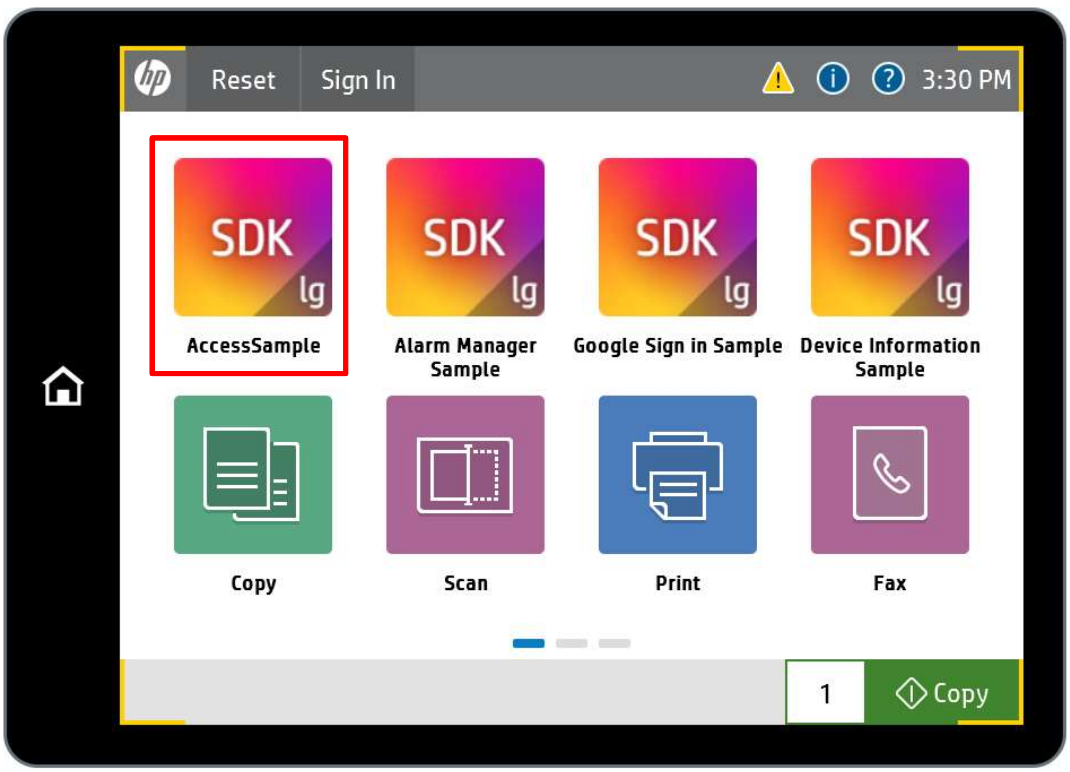
HPK Tool Pop-up for installing hpk file

9. Click "Install" within target device's host.



HPK Tool Pop-up for progressing installation

10. After installation is finished, click application's icon on a device.



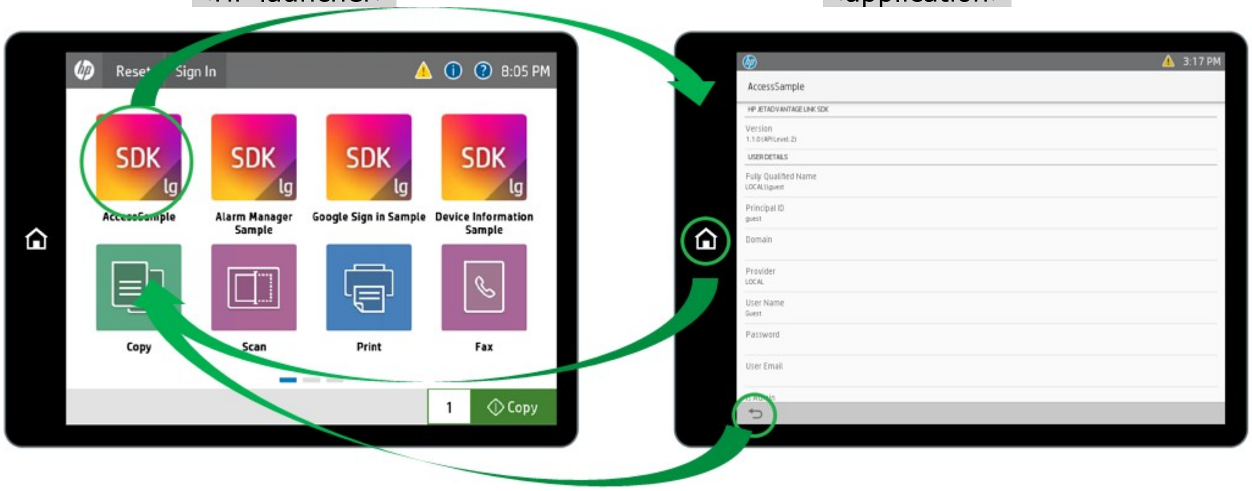
Main button to launch application

NOTE: You may need to press reset on the control panel before the button will appear. Once installed, you can launch your app from the front panel.

5.2.2.4 Step3-2. Enabling Enhanced Debug Experience Mode and installing .apk file

HP devices switch into Workpath when application is launched and switch back to HP launcher when application is closed. If developer enables Link Debug Bridge(LDB), developer can install application not packaged by hpk file using adb command, but developer will not be able to launch your application directly from HP launcher.

<HP launcher> <application>

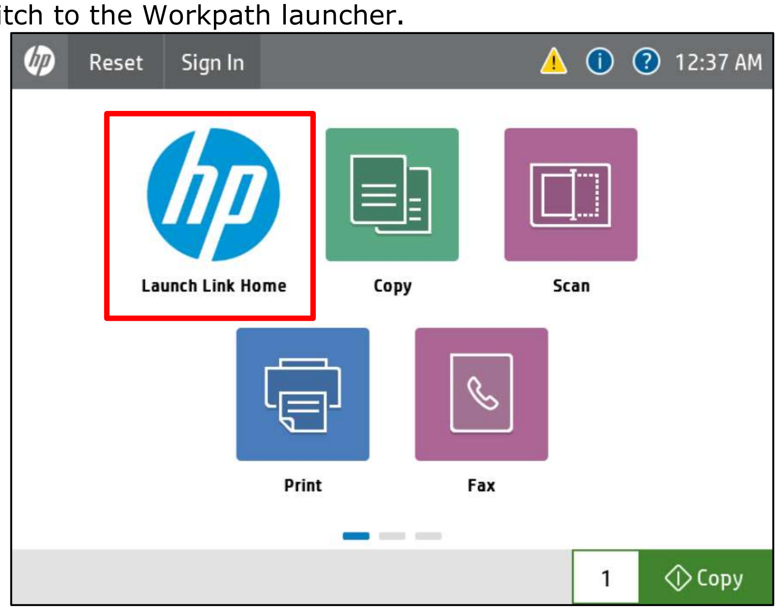


Switching between HP launcher and application

For developer's convenience, Workpath Platform provides an Enhanced Debug Experience (EDX) mode that enables developer to access the Workpath launcher just for application development. To turn on EDX mode, run the following command:

adb shell set_edx -v true

When EDX mode is enabled, after resetting a button will appear on the control panel to switch to the Workpath launcher.



The button to launch Workpath Home

To disable EDX mode, run the command:

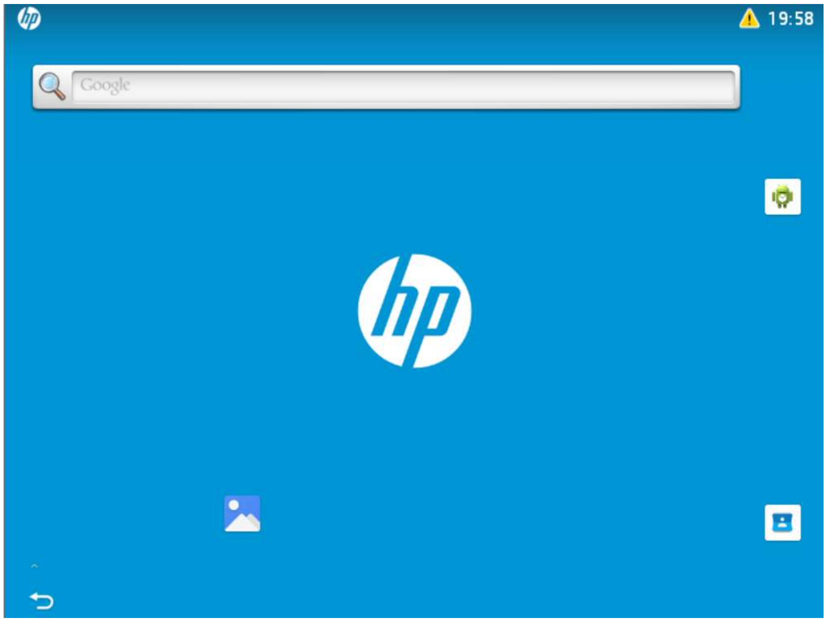
adb shell set_edx -v false

Following steps show how to enable EDX and install .apk for debug:

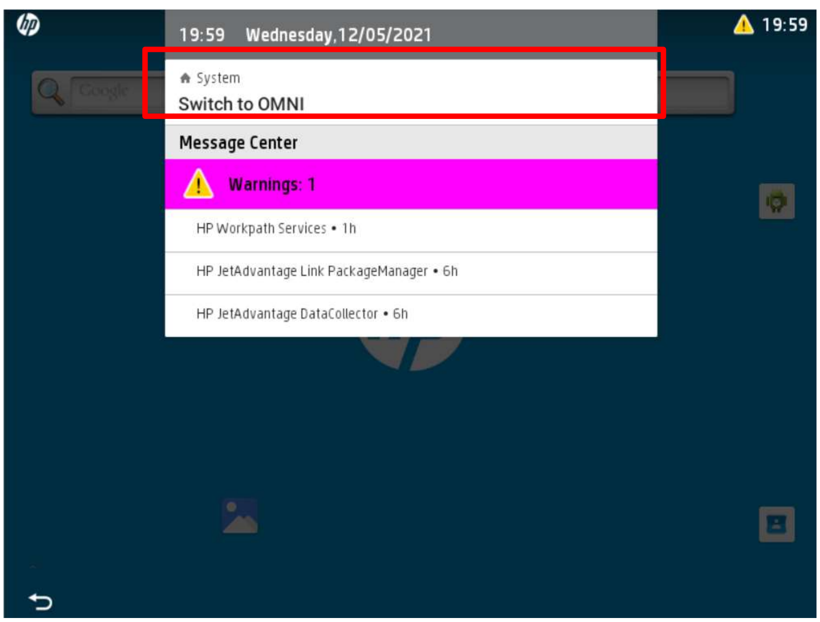
1. Enable EDX mode.
2. Install sample .apk using adb.

   adb connect <device_ip_address>

   adb install <path_to_sample_apk>
3. Click the "Launch Workpath Home" on HP launcher to switch into Workpath platform.



4. To switch back to HP launcher, scroll down the notification bar and click "Switch to OMNI".



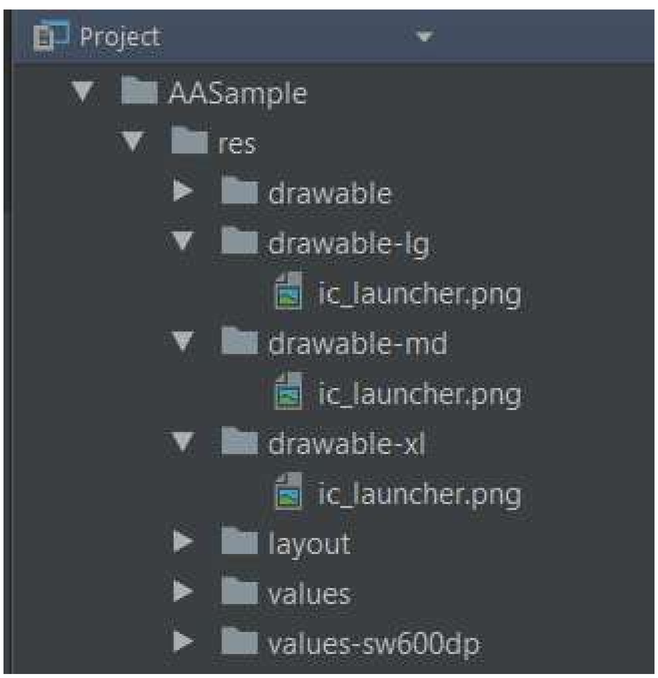
6 Tailoring sample application for HP Devices

HP device control panel sizes differ from panel sizes commonly found on mobile devices, so you may need to adjust your screen layouts to look their best on these panel sizes. In addition, there is a specific recommended HP launcher icon size for each control panel size.

The following table lists the HP device control panel sizes and their respective recommended HP launcher icon sizes:

Physical Panel Width	Resolution (WxH px)	HP Icon Size Category	Icon Size (WxH px)
10inch	1280x800	xl	179x179
8inch	1024x768	xl	179x179
8inch	800x600	lg	140x140
4.3inch	480x272	md	90x90

The below shows how to include HP launcher icons as HP icon size category-specific drawable in samples (e.g. res/drawable-lg/ic_hplauncher.png).



HP launcher icons in SDK samples

7 Dependency with SDK library

The Workpath SDK API of WorkpathLib.aar include open source libraries with specific version:

1) **simple-xml-2.7.1.jar**

If different version library is imported, you'll meet the confliction error.

Please exclude group to avoid confliction error. Here's the example :

```
dependencies {
implementation ('com.google.api-client:google-api-client:1.23.0'){
        exclude group: 'com.google.code.gson'
    }
```

Note: Starting from Workpath SDK API version 1.6.3, the embedded Gson library has been removed. If you were using Gson previously, you now need to import the Gson library manually.

8 Variation and Compatibility within an API

When developing applications against an API compatible device, remember that an API defines general behavior, while products or devices implement specific variants (geared to the product or device) of an API.

For full compatibility, applications must be prepared for the device to respond within the bounds of the API.

Products can vary in two different ways:

- Entire services may be omitted (e.g., the printer may not have a scanner)
- Variation within services is communicated by capabilities, either explicitly (via resource content), or implicitly (absence of a resource.)

In addition, full forward and backward compatibility within a series of releases is enabled. This means only "**compatible**" changes will be made, ensuring that existing clients will continue to work with implementations of newer API versions.

9 Android Platform APIs

HP Workpath platform is built on Android and supports standard Android Platform APIs, aside from limitations in security or hardware capabilities.

9.1 Android APIs with FutureSmart Integration

Some Android Platform APIs are additionally integrated with HP FutureSmart features and may provide specific behavior on FutureSmart devices.

9.1.1 Notifications

A notification is a message that Android displays outside of an app's UI to provide information to the user.

On FutureSmart devices, notifications posted via Android notification APIs are also displayed in the device Message Center. Notifications remain visible until the app cancels them. It is the app's responsibility to cancel a notification when it is no longer needed.

If the notification is configured to launch the app when tapped (by setting `NotificationCompat.Builder#setContentIntent`), the FutureSmart device Message Center displays an **Open** button for that notification to launch the app.

For details on the Android notification APIs, see:

- https://developer.android.com/reference/android/app/NotificationManager

- https://developer.android.com/reference/androidx/core/app/NotificationCompat.Builder

9.1.2 Wake Lock

A wake lock is an Android mechanism that allows an app to request that the device stay awake.

On FutureSmart devices, when an app acquires a wake lock using the Android `PowerManager.WakeLock` APIs, the device may remain awake and may not enter inactivity timeout or power-save mode until the wake lock is released. The app is responsible for releasing any wake lock it acquires.

For details on the Android `WakeLock` APIs, see:

- https://developer.android.com/reference/android/os/PowerManager.WakeLock

9.2 Android Restrictions and Tips

When targeting Android 12 (API level 31) or later, app developers should be aware of the following restrictions.

9.2.1 Restrictions on starting a foreground service from the background

Apps that target Android 12 (API level 31) or higher cannot start a foreground service while running in the background. One exception to this restriction is when the app triggers an exact alarm. For details on background start restrictions and scheduling alarms, see:

https://developer.android.com/develop/background-work/services/fgs/restrictions-bg-start

https://developer.android.com/training/scheduling/alarms

10 Legal notice

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11 Support

The HP Solution Business Program (SBP) support staff has established a procedure for submitting technical problems to the Incident Call Center (ICC). All SBP members must be registered with the Solution Programs Portal (SPP) to access technical support and other feedback or reporting activities.

Log on to the portal at <http://www.hp.com/go/solutions> to submit your issue to technical support. After you submit the issue, a report will be generated for the Incident Call Center. A Technical Support representative will be assigned to validate the incident and to work on a fix. HP will contact you for further information regarding the incident as quickly as possible.